Wide Character

Wide character is similar to char data type, except that wide char take up twice the space and can take on much larger values as a result. char can take 256 values which corresponds to entries in the ASCII table. On the other hand, wide char can take on 65536 values which corresponds to UNICODE values which is a recent international standard which allows for the encoding of characters for virtually all languages and commonly used symbols.

1.Just like the type for character constants is char, the type for wide character is wchar_t.

2.This data type occupies 2 or 4 bytes depending on the compiler being used.

3.Mostly the wchar_t datatype is used when international languages like Greek , Arabic ,Thai etc are used.

```
// An example C++ program to demonstrate use of wchar_t
#include <iostream>
using namespace std;
int main()
{
    wchar_t w  = L'A';
    cout << "Wide character value:: " << w << endl ;
    cout << "Size of the wide char is:: " << sizeof(w);
    return 0;
}
```

Output:

```
Wide character value:: 65

Size of the wide char is:: 2 or 4 depending upon compiler.
```

- L is the prefix for wide character literals and wide-character string literals which tells the compiler that that the char or string is of type wide-char.
- w is prefixed in operations like scanning (wcin) or printing (wcout) while operating wide-char type.

Wide char type array or string:

Below example demonstrate the wide-char type array

```
#include<iostream>

using namespace std;

int main() {

      char str1[] = "This is character array";

      cout << str1 << endl;

      wchar_t str2 [] = L"This is wide character array";

      wcout << str2;

}
```

Output

```
This is character array
This is wide character array
```

Functions for wide character array strings :

Function	Description
wcslen()	The syntax is size_t wcslen (const wchar_t* wcs); This function is used to get the length of the wide character string.
wcscat()	The syntax is: wchar_t *wcscat(wchar_t *strDest, const wchar_t *strSrc); This function is used to concatenate the source string with the destination string.
wcscpy()	The syntax is wchar_t *wcscpy(wchar_t *strDest, const wchar_t *strSrc); It helps to copy the source string to destination string.
wcsncpy()	The syntax is wchar_t* wcsncpy(wchar_t* dest, const wchar_t* src, size_t n); This function is used to copy first n characters of source to destination. When the end os source is less than n, then at the

Function	Description
	destination there will be some null characters.
wcscmp ()	The syntax is int wcscmp(const wchar_t* wcs1, const wchar_t* wcs2); This function is used to compare two wide character strings wcs1 and wcs2. It is like strcmp() function for normal string comparison.
wcsstr()	The syntax is const wchar_t* wcsstr (const wchar_t* wcs1, const wchar_t* wcs2); This function is used to find the first occurrence of wcs2 in wcs1. If it is not present, then it returns null
wcstok()	The syntax is wchar_t* wcstok(wchar_t* str, const wchar_t* delim, wchar_t ** ptr); This function is like strtok(). It helps to tokenize string which are created by using wide characters. It takes delimiter to tokenize the string.

Boolean Values:

A boolean variable is declared with the `bool` keyword and can only take the values `true` or `false`.

Example

```
#include<iostream>
using namespace std;
main()
{
        bool flag1=true;
        bool flag2=false
        cout<<flag1<<endl;
        cout<<flag2<<endl;
}
```

Output:

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1

0

- We can use bool type variables or values true and false in mathematical expressions also.For instance,

```
int x = false + true + 6;
```

is valid and the expression on right will evaluate to 7 as false has value 0 and true will have value 1.

- It is also possible to convert implicitly the data type integers or floating point values to bool type. For example, the statements-

bool flag1=0 ; // false

bool flag2=23.4 ; // true

bool flag3= 100; // true

Example :

```
#include<iostream>

using namespace std;
int main()
{
    int x1 = 10, x2 = 20, m = 2;
    bool b1, b2;
    b1 = x1 == x2; // false
    b2 = x1 < x2; // true
    cout << "b1 is = " << b1 << "\n";
    cout << "b2 is = " << b2 << "\n";
    bool b3 = true;
    if (b3)
        cout << "Yes" << "\n";
    else
        cout << "No" << "\n";
    int x3 = false + 5 * m - b3;
    cout << x3;
return 0;
}
```

Output:

```
b1 is = 0

b2 is = 1

Yes

9
```

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